# Fundamental Trigonometric Identities

## Section 1: Reciprocal and Quotient Identities

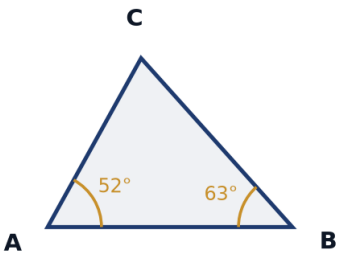
**1** Simplify the expression $\tan x \cdot \csc x$.

**a.** $\sec x$

**b.** $\cos x$

**c.** $\sin x$

**d.** $\cot x$




**2** Simplify $\frac{\cos x}{\cot x}$.

**a.** $\sin x$

**b.** $\cos^2 x$

**c.** $\tan x$

**d.** $\csc x$

**3** If $\sin\theta = -3/5$ and $\tan\theta > 0$, find $\cos\theta$.

**a.** $-4/5$

**b.** $4/5$

**c.** $-3/4$

**d.** $5/4$

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## Section 2: Pythagorean Identities

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**4** Which identity is directly derived by dividing $\sin^2\theta + \cos^2\theta = 1$ by $\cos^2\theta$?

**a.** $1 + \tan^2\theta = \sec^2\theta$

**b.** $1 + \cot^2\theta = \csc^2\theta$

**c.** $\sin^2\theta = 1 - \cos^2\theta$

**d.** $\tan^2\theta + \cot^2\theta = 1$

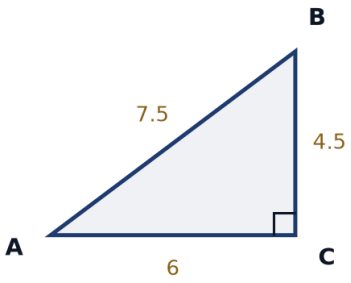
**5** Simplify $(\sec\theta - 1)(\sec\theta + 1)$.

**a.** $\tan^2\theta$

**b.** $\cot^2\theta$

**c.** $\cos^2\theta$

**d.** $1$




**6** Simplify $\frac{1 - \cos^2 x}{\sin x}$.

**a.** $\sin x$

**b.** $\cos x$

**c.** $\csc x$

**d.** $1$

## Section 3: Even/Odd Identities and Symmetry

**7** Which trigonometric functions are even functions ($f(-\theta) = f(\theta)$)?

**a.** $\cos\theta$ and $\sec\theta$

**b.** $\sin\theta$ and $\csc\theta$

**c.** $\tan\theta$ and $\cot\theta$

**d.** All six functions

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**8** Simplify $\sin(-\theta) \cdot \cot(-\theta)$.

**a.** $\cos\theta$

**b.** $-\cos\theta$

**c.** $\sin\theta$

**d.** $-\sin\theta$

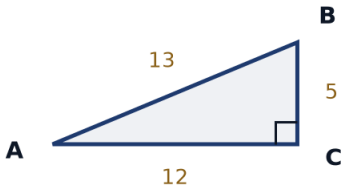
**9** Simplify $\frac{\tan(-x)}{\sin(-x)}$.

**a.** $\sec x$

**b.** $-\sec x$

**c.** $\cos x$

**d.** $-\cos x$




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## Section 4: Verifying Identities Algebraically

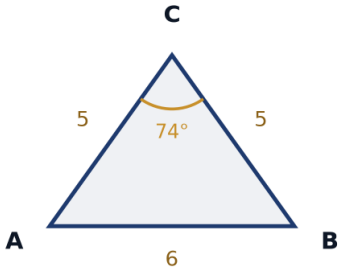
**10** Prove $\frac{\sin x}{1+\cos x} = \frac{1-\cos x}{\sin x}$. What technique verifies this algebraically?

**a.** Multiply numerator and denominator of the left side by the conjugate $(1-\cos x)$

**b.** Square both sides

**c.** Take the derivative

**d.** Cross multiply and set to 0




**11** Simplify $\frac{\sec x - \cos x}{\sin x}$.

**a.** $\tan x$

**b.** $\cot x$

**c.** $\sec x$

**d.** $\sin x$

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**12** Simplify $(1 + \tan^2\theta)(1 - \sin^2\theta)$.

**a.** $1$

**b.** $\sec^2\theta$

**c.** $\tan^2\theta$

**d.** $\cos^2\theta$

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